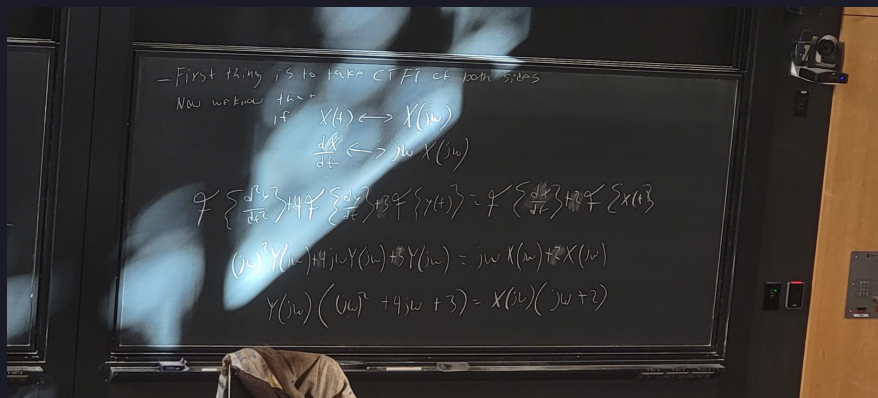
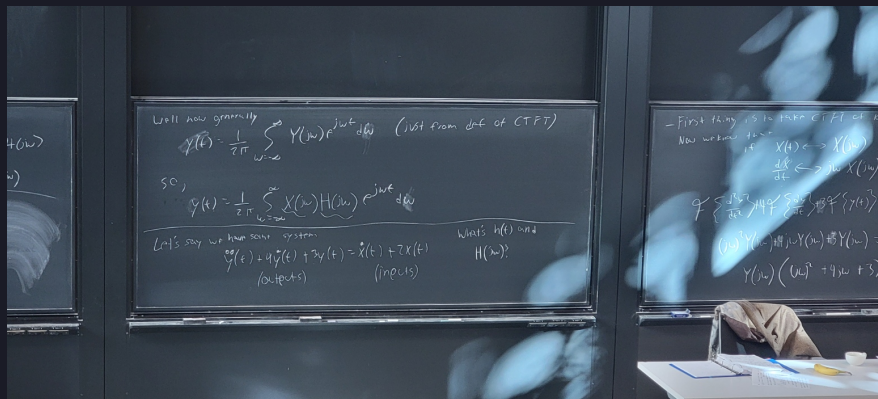


Convolution in Fourier

1 Continuous



Then we get $H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}$

The inverse Fourier transform of this is $h(t)$. This is most easily done with the Fourier transform pairs that we derived.

We use partial fractions to separate out

$$H(j\omega) = \frac{\frac{1}{2}}{j\omega + 1} + \frac{\frac{1}{2}}{j\omega + 3} \quad (1)$$

Recall

$$e^{-at}u(t) \leftrightarrow \frac{1}{a+j\omega}$$

$$\text{so } h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t)$$

Lets take an arbitrary input $x(t) = e^{-4t}$. The Fourier transform is $X(j\omega) = \frac{1}{4+j\omega}$

Then we have

$$Y(j\omega) = H(j\omega)X(j\omega)$$

and then go back to the time domain with the Fourier pairs.

One more interesting property:

Multiplication and convolution are DUALS

$$\begin{aligned} x(t) * y(t) &\leftrightarrow X(j\omega)Y(j\omega) \\ x(t)y(t) &\leftrightarrow \frac{1}{2\pi}X(j\omega) * Y(j\omega) \end{aligned}$$

Note that if the Fourier domain is a bunch of delta functions that are far enough apart, then we just get multiple copies of the original function.

2 Discrete

Lets take an aperiodic signal $x[n]$

We truncate it to only exist over one period N_1 , and zero elsewhere, then make it a periodic function by making copies at a period $N > 2 * N_1$

call this $\tilde{x}[n]$ with period N and $\hat{\omega}_0 = \frac{2\pi}{N}$.

We now have the Fourier series representation

$$a_k = \frac{1}{N} \sum_N \tilde{x}[n] e^{-jk(\frac{2\pi}{N})n} \text{ and } \tilde{x} = \sum_N a_k e^{jk(\frac{2\pi}{N})n}.$$

This is the same as

$$a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(\frac{2\pi}{N})n} \quad (2)$$

We define

$$X(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{\omega}n}, \text{ so} \quad (3)$$

$$a_k = \frac{1}{N} X(e^{jk\hat{\omega}_0}) \quad (4)$$